TCP – Congestion Control Algorithms
Network programming in python

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How to run the program:

We'll be using a Linux based operating system.
First, you'll need to download the following files and put them in the same directory:

1. Sender.py
2. Receiver.py

Then, open the integrated terminal ad run the following commands:

1. Python3 Receiver.py

Open a second terminal and run the following commands:

1. Python3 Sender.py

Once the connection is completed and the sending of the file has started, you'll asked if you'll want to resend the file 'Y' means yes, and 'N' means no.
To terminate the connection type 'N'.

Note: when checking for packet loss install the TC tool using the following command:
sudo apt install iproute
to see the packet loss, use the following command in the first terminal before the command for 'Receiver.py':
sudo tc qdisc change dev lo root netem loss XX%.

About the system:

In this project, we were instructed to write two program files Sender.py and Receiver.py which operate in the client server method (it's a model in networking computing system, which defines the relationship between cooperating programs. The server is a passive software that listens to the network and waits to receive requests. The client on the other hand, is activated by the user and turns to the server when he needs information or services from it).

In our case, the sender will send a message, a file sized at least 2MB. And the receiver will measure the time it took to get the said file.
The file will be sent in two halfs, each part will be sent according to the CC algorithms learned in class.
The first half will be sent using the RENO algorithm, while the second half will be sent using the CUBIC algorithm.

In addition, we will use a packet loos tool called TC, and will monitor the results using Wireshark.

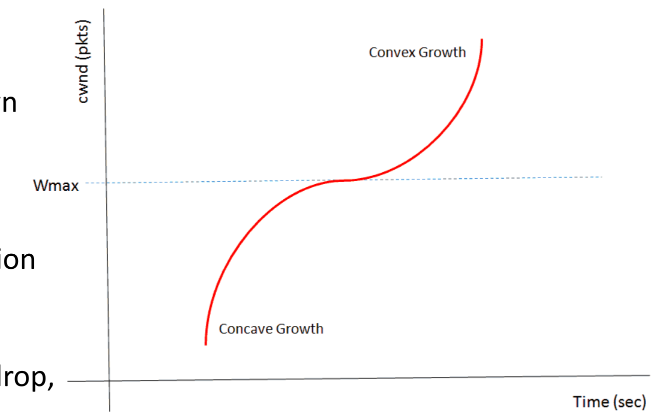
About the CC algorithms:

congestion on the internet happens when the computers send more packets to a particular link than that link can handle. If a link is receiving more packets than can fit on the wire, it will begin queueing up those packets, and if the link's queues fill up, the link will begin dropping those packets. Therefore we have the CC - Algorithm. In this program we use the reno and cubic CC Algorithms.

Cubic:

CUBIC is a standard TCP congestion control algorithm that uses a cubic function on the sender side to improve scalability and stability over fast and long-distance networks.

As we can see, the cubic function is very powerful- as x Is lower the function grows very quickly, and then slows down As it approaches a particular point, and after crossing that Specific point the function grows fairly quickly again.



The congestion window that has grown from the cubic function has the following properties:

1. Starts growing fast.
2. as the algorithm approaches the window at the last drop, the congestion window begins growing more slowly.
3. If the congestion window gets to the point at which drops occurred the last time, and does not experience a drop, begin growing slowly again, but then increase more quickly.

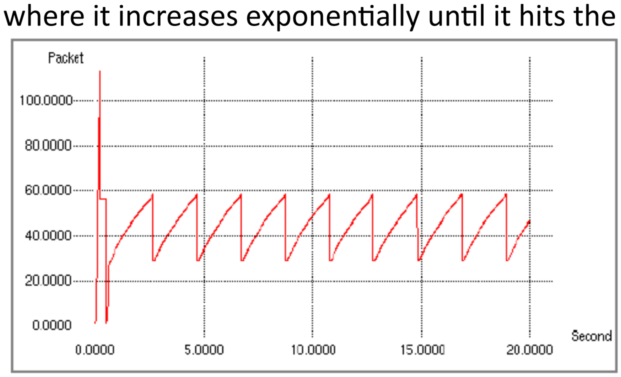
During the concave phase, the congestion window is catching up to where a packet was lost last time and slows down its growth to be fair to traditional TCP senders. If the algorithm gets there without experiencing drops, it can move into an "exploratory" phase, in which it grows quickly to discover newly available bandwidth.

Reno:

this algorithm operates by increasing the congestion window size exponentially until some threshold is reached.

After that threshold is reached, the function increases linearly. Once it experience drops, it will aggressively drop the window size, and then enter "slow start" mode again, where it increases exponentially until it hits the threshold. Once the threshold is hit, it will increase linearly once again.

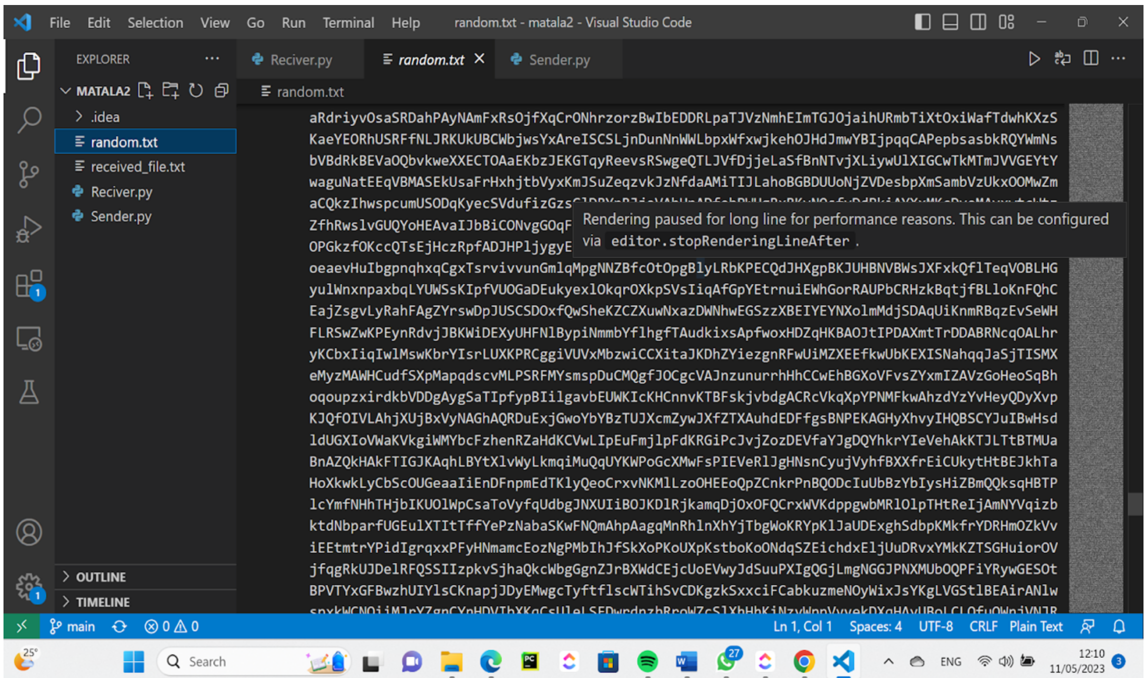
Since Reno grows linearly after crossing the slow start threshold, if a lot more bandwidths become available on a link, it'll take a long time for the algorithm to "discover" that available bandwidth.



Our code:

The main goal of our code is to create two objects of the classes sender and receiver. A terminal running the receiver file will run a short main program initialising a receiver object that acts as the server in the client server model. We then run a terminal that runs the sender file acting as the client in the server client model.

The server object is initialised and creates a random file of size between 3-6MB of chars. A file is created of random chars each line with 50 chars. As shown below:



The receiver object creates a socket binds and listens.

Reciver.py:

we create a receiver object with a file name (for where to save what it receives), host , port and our id’s for sending acks.

The instance of our receiver object then runs it function run() with a flag 0 indicating this is the first time its been made.

```python
def main():

    filename = "received_file.txt"
    host = "127.0.0.1"  # self ip
    port = 9999 # port 9999 free

    r1 = Receiver(filename, host, port, '332307073', '214329633')
    r1.run(0)


if __name__ == "__main__":
    main()
```

the constructor of our receiver class has extra fields such as a log for saving times, auth the XOR of our id’s for sending acks and more variables that will help us later down the line.

```python
class Receiver:
    def __init__(self, filename, host, port, id_1, id_2):
        self.filename = filename
        self.host = host
        self.port = port
        self.chunk_size = 1024
        #log for statistics
        self.log = []
        #ids for XOR auth operation
        self.xor = bin(int(id_1)^int(id_2))[2:]
        self.auth = str(self.xor).encode()
        self.conn = None
        self.addr = None
        self.filesize = 0
        self.s = None
        self.max = 0
        self.ended = 0
```

The run function opens a file to write into. If the flag is 0 meaning this is the first time we initialise a socket and save it to self.s and bind and listen we accept the connection and save the variables, if flag is not 0 we will later on see why we do not initialise the socket again.

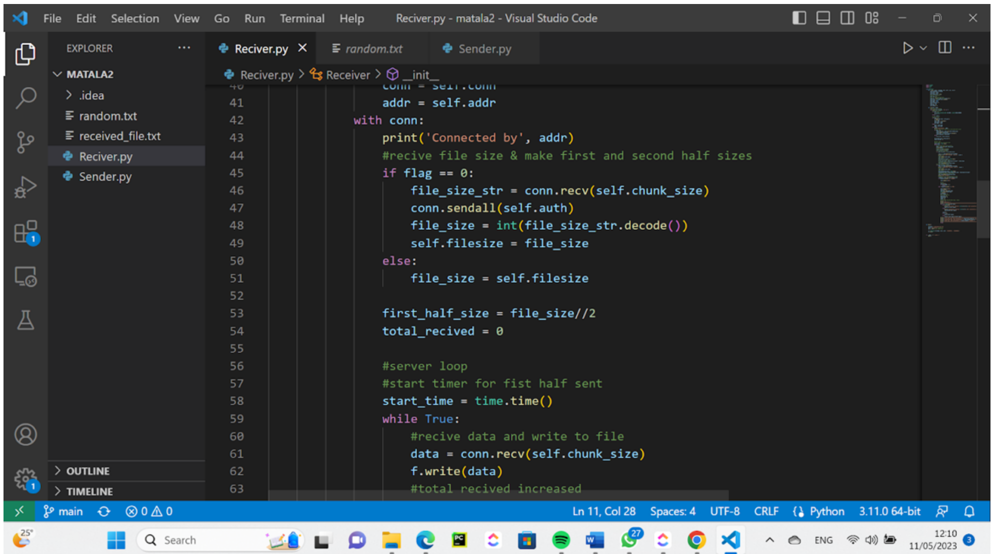
```python
        self.ended = 0

    def run(self, flag):
        #open file for writing
        with open(self.filename, 'wb') as f:
            #initilize socket
            if flag == 0:
                s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
                self.s = s
            else:
                s = self.s
            #bind & listen
            if flag == 0:
                s.bind((self.host, self.port))
                s.listen()
                conn, addr = s.accept()
                self.conn = conn
                self.addr = addr
            else:
                conn = self.conn
                addr = self.addr
            with conn:
                print('Connected by', addr)
```

With the connection we print connected to the address of the sender.

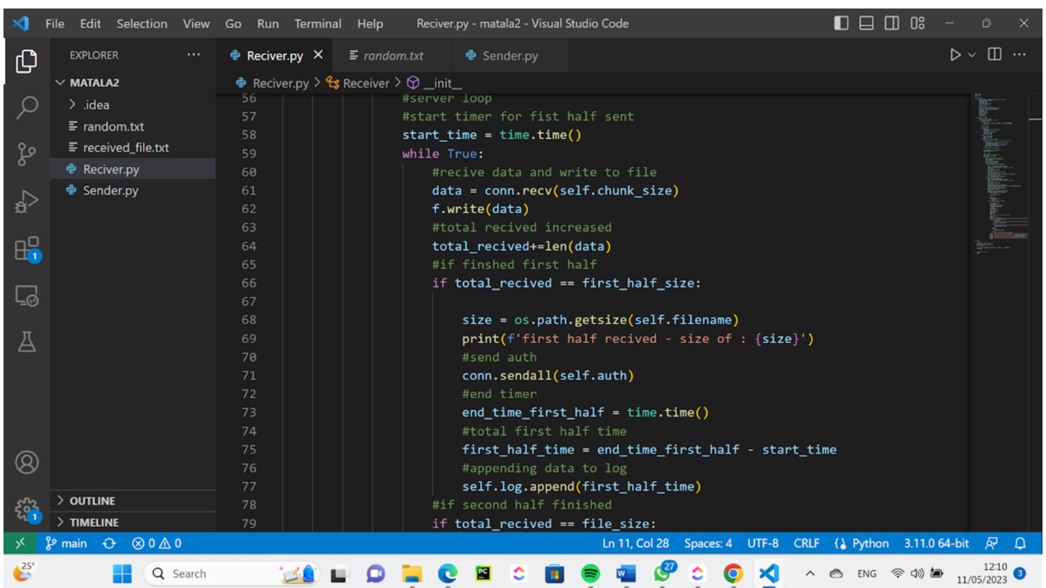
If this is our first time (flag = 0) the sender sends us the length of the file and we ack back and save it.

We then save the first_half size fo the file and keep track of total sent starting at 0.

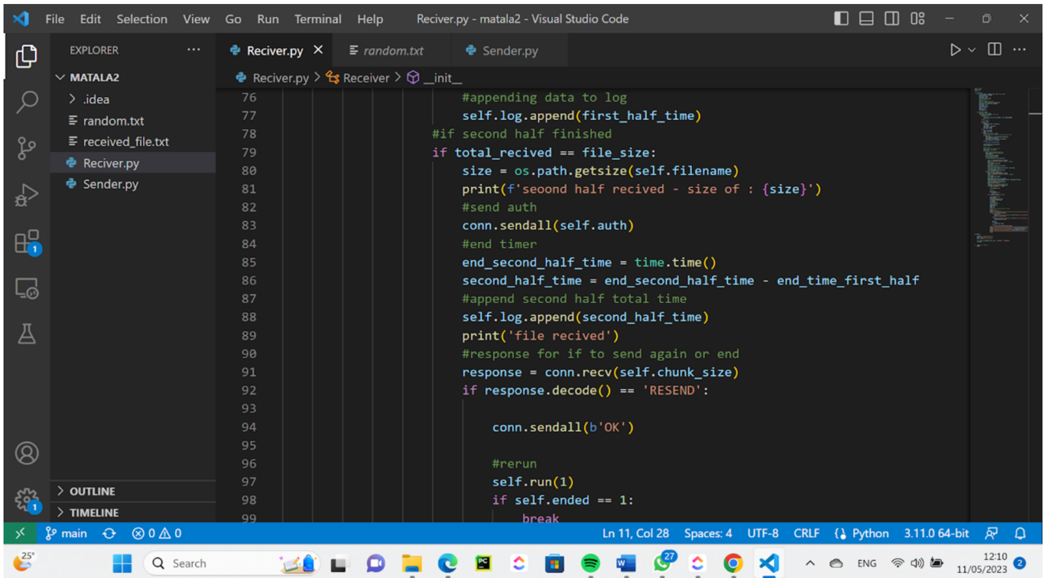
Start a timer and enter the while loop.



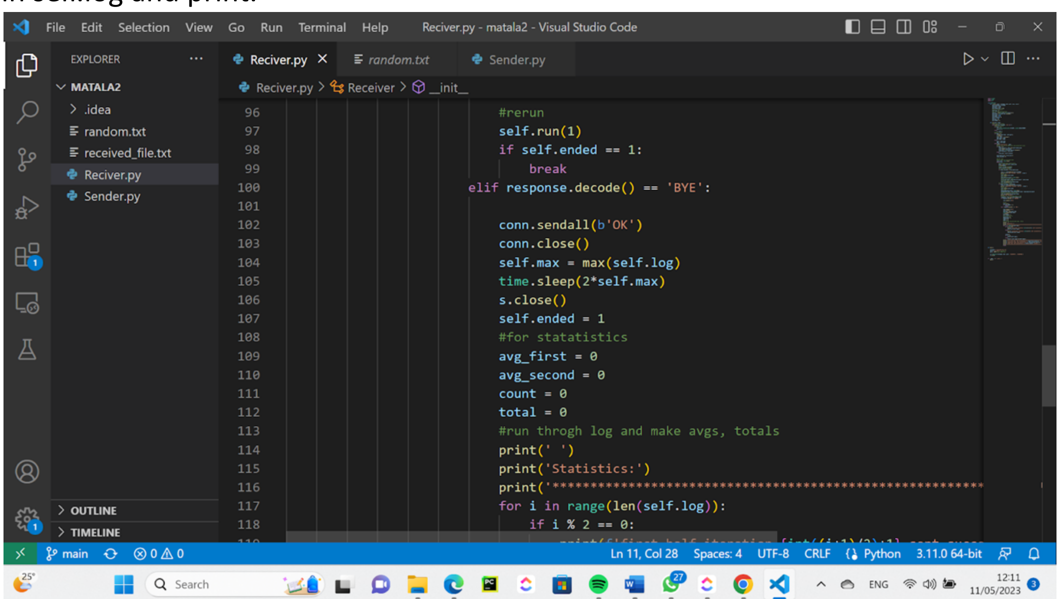
In the while loop we receive chunks of the data and write it to the file, we update the total amount of data received. If the total amount received equals the first half size we send a ack and stop the timer and save the time to the log.

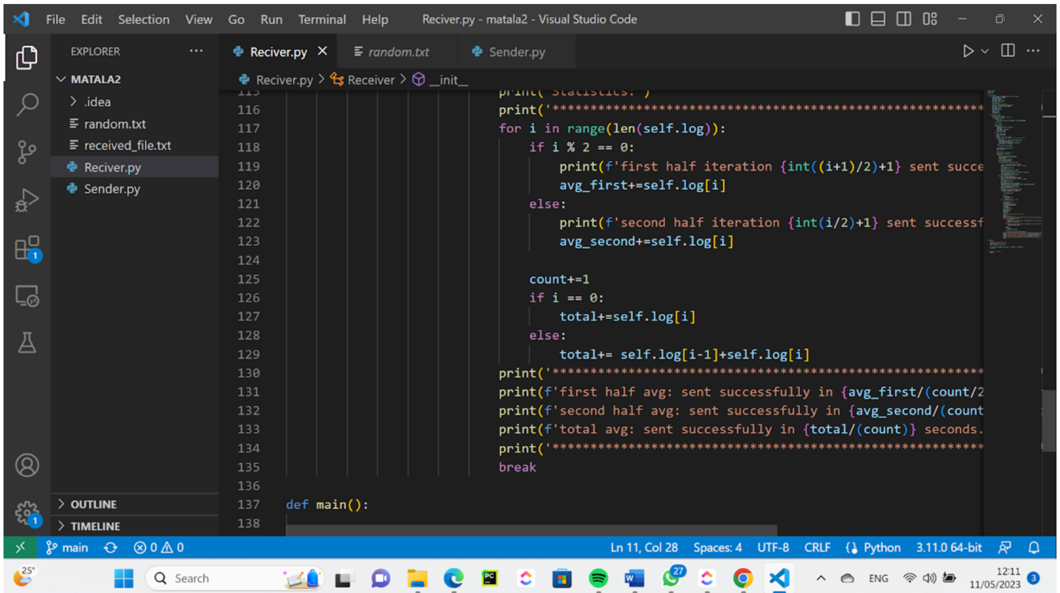


If the total amount received equals the size of the file, we send an ack and stop the timer. We append to the log the total time it took to receive the second half. And print file received. We then wait for the response of the sender if they want to resend or exit. If the decoded response is resend we recursively call the run function with flag 1. This will keep the connection open and work. Notice after we exit the recursive function we have another flag if it is 1 we break the while loop. This will allow us to exit gracefully, when closing the connection and closing the socket in the recursions.



If the response decoded is bye, we ack and close the connection. We then wait 2*rtt in our case we took the longest sending time of a half of a file to be careful. We then change the objects ended = 1 so that when we come back from the recursive functions we exit gracefully. (as opposed to being in a while loop trying to receive data from a closed connection and socket). We then compute the averages based on the values stored in self.log and print.



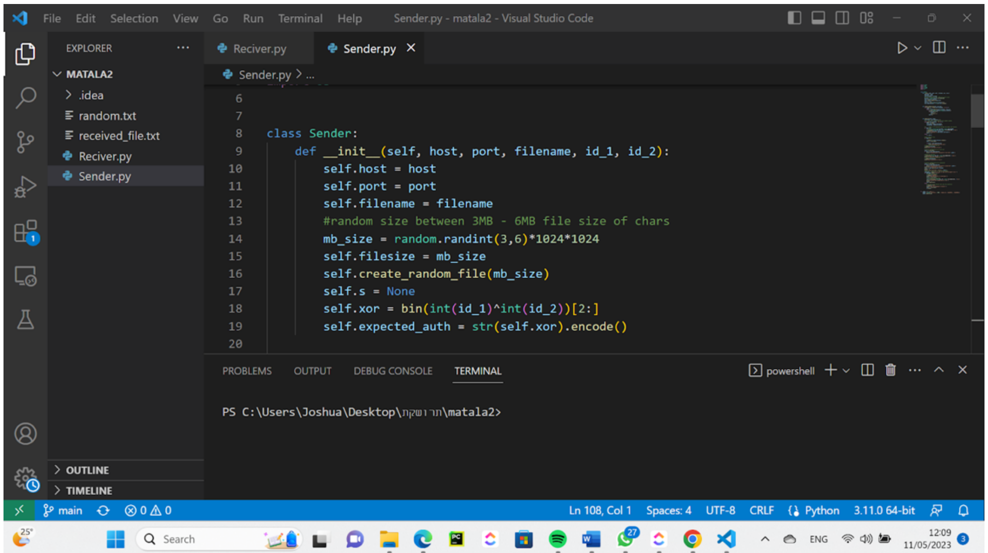

File Edit Selection View Go Run Terminal Help
Reciver.py - matala2 - Visual Studio Code
EXPLORER
MATALA2
.idea
random.txt
received_file.txt
Reciver.py
Sender.py
Reciver.py
random.txt
Sender.py
Reciver.py > Receiver > __init__
print('***
for i in range(len(self.log)):
if i % 2 == 0:
print(f'first half iteration {int((i+1)/2)+1} sent succe
avg_first+=self.log[i]
else:
print(f'second half iteration {int(i/2)+1} sent successf
avg_second+=self.log[i]
count+=1
if i == 0:
total+=self.log[i]
else:
total+= self.log[i-1]+self.log[i]
print('***
print(f'first half avg: sent successfully in {avg_first/(count/2
print(f'second half avg: sent successfully in {avg_second/(count
print(f'total avg: sent successfully in {total/(count)} seconds.
print('***
break
def main():
OUTLINE
TIMELINE
main
Ln 11, Col 28
Spaces: 4
UTF-8
CRLF
Python
3.11.0 64-bit
Search
ENG
12:11
11/05/2023

Sender.py:

The main is run and a sender object is created with a host, port , filename and our id's, then its send_file function is run with a flag = 0.

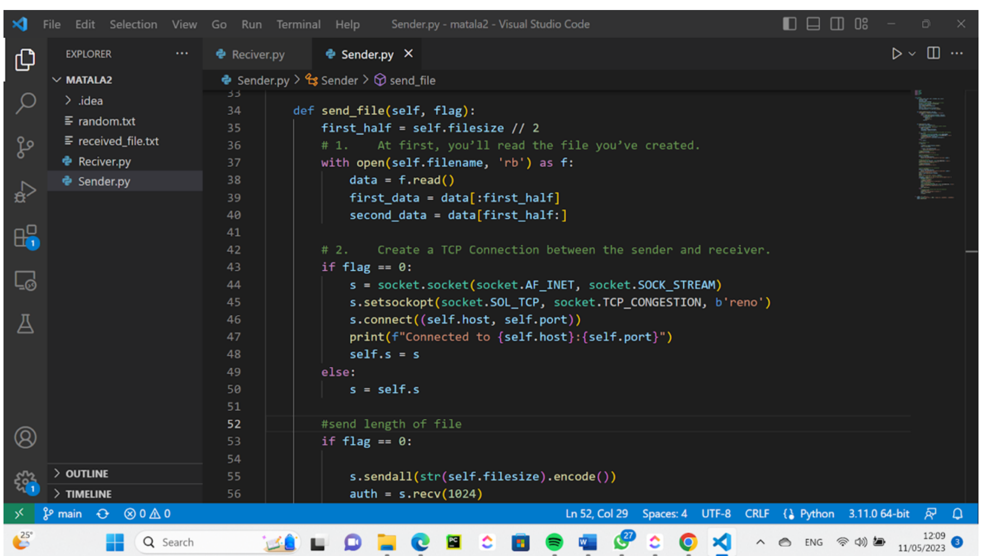
```python
if __name__ == "__main__":
    sender = Sender('127.0.0.1', 9999, 'random.txt','332307073', '214329633')
    sender.send_file(0)
```

The sender class constructor creates the object and creates a random size between 3-6MB then class create random file function.

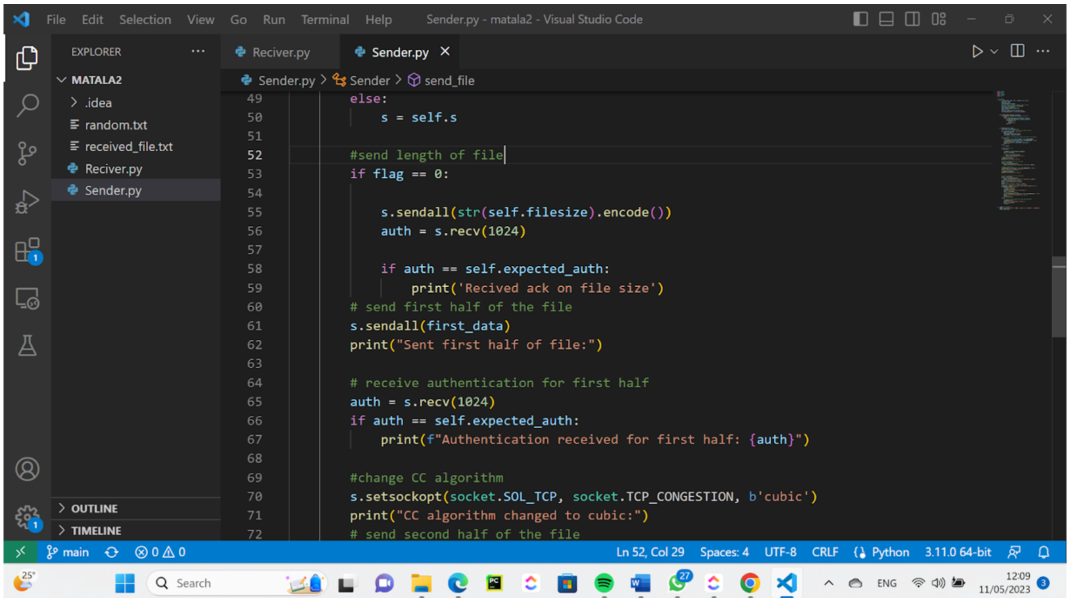


The create random file function creates a random file the of size between 3-6MB and writes chars in lines of 50.

The sendfile function the is called with a flag 0. It opens the written file and reads the data and saves it in two halves to be sent. If the flag is 0 it creates a socket and connects to the receiver and saves the socket



If the flag is 0 send the length of the file to the receiver. Then send the first half and wait for ack from receiver. Print first half sent and change CC algo to cubic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EXPLORER
MATALA2
.idea
random.txt
received_file.txt
Reciver.py
Sender.py
Reciver.py
Sender.py
Sender.py > Sender > send_file
else:
s = self.s
#send length of file
if flag == 0:
s.sendall(str(self.filesize).encode())
auth = s.recv(1024)
if auth == self.expected_auth:
print('Recived ack on file size')
# send first half of the file
s.sendall(first_data)
print("Sent first half of file:")
# receive authentication for first half
auth = s.recv(1024)
if auth == self.expected_auth:
print(f"Authentication received for first half: {auth}")
#change CC algorithm
s.setsockopt(socket.SOL_TCP, socket.TCP_CONGESTION, b'cubic')
print("CC algorithm changed to cubic:")
# send second half of the file
OUTLINE
TIMELINE
main
Ln 52, Col 29 Spaces: 4 UTF-8 CRLF Python 3.11.0 64-bit
Search
ENG
12:09
11/05/2023

Our findings running different levels of packet loss:

<u>0% loss:</u>

For 0 percent loss we sent the file once as shown

```
● jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc add dev lo root netem loss 0%
  [sudo] password for jodogo:
○ jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
  Connected to 127.0.0.1:9999
  Recived ack on file size
  Sent first half of file:
  Authentication received for first half: b'10001110000101100001010'
  CC algorithm changed to cubic:
  Sent second half of file:
  Authentication received for second half: b'10001110000101100001010'
  Sent second half of file:

  total file sent - size of : 6291456
  Do you want to send the file again? (y/n)
```

The receiver received the file for the first time as shown:

```
○ jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
  Connected by ('127.0.0.1', 46184)
  first half recived - size of : 3141632
  seoond half recived - size of : 6287360
  file recived
```

We then sent the file again as shown:

```
● jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc add dev lo root netem loss 0%
  [sudo] password for jodogo:
○ jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
  Connected to 127.0.0.1:9999
  Recived ack on file size
  Sent first half of file:
  Authentication received for first half: b'10001110000101100001010'
  CC algorithm changed to cubic:
  Sent second half of file:
  Authentication received for second half: b'10001110000101100001010'
  Sent second half of file:

  total file sent - size of : 6291456
  Do you want to send the file again? (y/n) y
  Notified receiver to send again.
  CC algorithm changed back to reno.

  Sent first half of file:
  Authentication received for first half: b'10001110000101100001010'
  CC algorithm changed to cubic:
  Sent second half of file:
  Authentication received for second half: b'10001110000101100001010'
  Sent second half of file:

  total file sent - size of : 6291456
  Do you want to send the file again? (y/n)
```

The receiver received the file again for the second time as shown:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
Connected by ('127.0.0.1', 46184)
first half recived - size of : 3141632
seoond half recived - size of : 6287360
file recived
Connected by ('127.0.0.1', 46184)
first half recived - size of : 3141632
seoond half recived - size of : 6287360
file recived
```

We then sent the file again multiple times (more than or at least 5 times) and then exited

```
total file sent - size of : 6291456
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n) n
Sent exit message to receiver.
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

The receiver received all the times the file was sent and printed this analysis: as one can see this is extremely fast and not surprising as there is 0% packet loss.

```
file recived

Statistics:
****************************************************************************
first half iteration 1 sent successfully in 0.007754325866699219 seconds.
second half iteration 1 sent successfully in 0.006215810775756836 seconds.
first half iteration 2 sent successfully in 0.03510260581970215 seconds.
second half iteration 2 sent successfully in 0.007299184799194336 seconds.
first half iteration 3 sent successfully in 0.01335120201108398 seconds.
second half iteration 3 sent successfully in 0.01095890998840332 seconds.
first half iteration 4 sent successfully in 0.010681390762329102 seconds.
second half iteration 4 sent successfully in 0.006400108337402344 seconds.
first half iteration 5 sent successfully in 0.012102603912353516 seconds.
second half iteration 5 sent successfully in 0.014298677444458008 seconds.
first half iteration 6 sent successfully in 0.01311182975769043 seconds.
second half iteration 6 sent successfully in 0.005118846893310547 seconds.
first half iteration 7 sent successfully in 0.009816646575927734 seconds.
second half iteration 7 sent successfully in 0.011287927627563477 seconds.
first half iteration 8 sent successfully in 0.009810209274291992 seconds.
second half iteration 8 sent successfully in 0.00483393669128418 seconds.
first half iteration 9 sent successfully in 0.010222196578979492 seconds.
second half iteration 9 sent successfully in 0.01852726936340332 seconds.
****************************************************************************
first half avg: sent successfully in 0.01355033450656467 seconds.
second half avg: sent successfully in 0.009437852435641818 seconds.
total avg: sent successfully in 0.021958894199795194 seconds.
****************************************************************************
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

We shall do this process for the following levels of loss: 10%, 15%, 20%, 25%, 30%

10% loss:

Sending file for the first time

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc change dev lo root netem loss 10%
[sudo] password for jodogo:
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
Connected to 127.0.0.1:9999
Recived ack on file size
Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 3145728
Do you want to send the file again? (y/n)
```

Receiving file for the first time

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
Connected by ('127.0.0.1', 52500)
first half recived - size of : 1568768
seoond half recived - size of : 3144986
file recived
```

Finished sending file multiple times to receiver

```
total file sent - size of : 3145728
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 3145728
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 3145728
Do you want to send the file again? (y/n) n
Sent exit message to receiver.
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

Receiver printing stats: as we can see the process took slightly longer but still bearable.

```
file recived

Statistics:
**************************************************************************
first half iteration 1 sent successfully in 0.2158799171447754 seconds.
second half iteration 1 sent successfully in 1.121300220489502 seconds.
first half iteration 2 sent successfully in 0.08398771286010742 seconds.
second half iteration 2 sent successfully in 0.027344703674316406 seconds.
first half iteration 3 sent successfully in 0.03571462631225586 seconds.
second half iteration 3 sent successfully in 0.003339052200317383 seconds.
first half iteration 4 sent successfully in 0.21660518646240234 seconds.
second half iteration 4 sent successfully in 0.21417617797851562 seconds.
first half iteration 5 sent successfully in 0.048050642013549805 seconds.
second half iteration 5 sent successfully in 0.0029294490814208984 seconds.
first half iteration 6 sent successfully in 0.13849902153015137 seconds.
second half iteration 6 sent successfully in 0.21997570991516113 seconds.
first half iteration 7 sent successfully in 0.005781650543212891 seconds.
second half iteration 7 sent successfully in 0.20874285697937012 seconds.
first half iteration 8 sent successfully in 0.2299962043762207 seconds.
second half iteration 8 sent successfully in 0.005125761032104492 seconds.
first half iteration 9 sent successfully in 0.006598234176635742 seconds.
second half iteration 9 sent successfully in 0.004861593246459961 seconds.
**************************************************************************
first half avg: sent successfully in 0.10901257726881239 seconds.
second half avg: sent successfully in 0.20086616939968532 seconds.
total avg: sent successfully in 0.3096086581548055 seconds.
**************************************************************************
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

<u>15% loss:</u>

Sending file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc change dev lo root netem loss 15%
[sudo] password for jodogo:
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
Connected to 127.0.0.1:9999
Recived ack on file size
Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 3145728
Do you want to send the file again? (y/n)
```

Receiving file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
Connected by ('127.0.0.1', 60580)
first half recived - size of : 1568768
seoond half recived - size of : 3141632
file recived
```

Finished sending file multiple times to receiver:

```
total file sent - size of : 3145728
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 3145728
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 3145728
Do you want to send the file again? (y/n) n
Sent exit message to receiver.
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

Receiver printing stats: here also the time has gotten slightly longer

We can see the first half sent via reno CC algo on average is doing better than second half using cubic CC algo

```
file recived

Statistics:
**************************************************************************
first half iteration 1 sent successfully in 0.029301166534423828 seconds.
second half iteration 1 sent successfully in 0.019957780838012695 seconds.
first half iteration 2 sent successfully in 0.2243657112121582 seconds.
second half iteration 2 sent successfully in 0.48931217193603516 seconds.
first half iteration 3 sent successfully in 1.7009327411651611 seconds.
second half iteration 3 sent successfully in 1.8063304424285889 seconds.
first half iteration 4 sent successfully in 0.0961141586303711 seconds.
second half iteration 4 sent successfully in 0.0034341812133789062 seconds.
first half iteration 5 sent successfully in 0.11625838279724121 seconds.
second half iteration 5 sent successfully in 0.424685001373291 seconds.
first half iteration 6 sent successfully in 0.5196244716644287 seconds.
second half iteration 6 sent successfully in 0.3408691883087158 seconds.
first half iteration 7 sent successfully in 0.23624181747436523 seconds.
second half iteration 7 sent successfully in 0.6354207992553711 seconds.
first half iteration 8 sent successfully in 0.11029577255249023 seconds.
second half iteration 8 sent successfully in 0.009691953659057617 seconds.
first half iteration 9 sent successfully in 0.008368730545043945 seconds.
second half iteration 9 sent successfully in 0.9380323886871338 seconds.
**************************************************************************
first half avg: sent successfully in 0.33794477250840926 seconds.
second half avg: sent successfully in 0.5186371008555094 seconds.
total avg: sent successfully in 0.8044689628813002 seconds.
**************************************************************************
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

20% loss:

Sending file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc change dev lo root netem loss 20%
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
Connected to 127.0.0.1:9999
Recived ack on file size
Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 4194304
Do you want to send the file again? (y/n)
```

Receiving file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
Connected by ('127.0.0.1', 33180)
first half recived - size of : 2093056
seoond half recived - size of : 4192714
file recived
```

Finished sending file to receiver multiple times:

```
total file sent - size of : 4194304
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 4194304
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 4194304
Do you want to send the file again? (y/n) n
Sent exit message to receiver.
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

Receiver prints stats: we can see here that a big change has been made as now the files take seconds to send rather than milliseconds, notice we encounter outliners such as in the second last row of 19 seconds. We should consider taking this in to account when doing the averages.

```
first half recived - size of : 2096092
seoond half recived - size of : 4190878
file recived

Statistics:
*************************************************************************
first half iteration 1 sent successfully in 4.7404820919036865 seconds.
second half iteration 1 sent successfully in 4.281578779220581 seconds.
first half iteration 2 sent successfully in 1.3726799488067627 seconds.
second half iteration 2 sent successfully in 0.4253201484680176 seconds.
first half iteration 3 sent successfully in 6.470535516738892 seconds.
second half iteration 3 sent successfully in 1.2220697402954102 seconds.
first half iteration 4 sent successfully in 0.2625274658203125 seconds.
second half iteration 4 sent successfully in 4.36931586265564 seconds.
first half iteration 5 sent successfully in 0.48596882820129395 seconds.
second half iteration 5 sent successfully in 0.752655029296875 seconds.
first half iteration 6 sent successfully in 4.721026420593262 seconds.
second half iteration 6 sent successfully in 1.1812093257904053 seconds.
first half iteration 7 sent successfully in 0.2794656753540039 seconds.
second half iteration 7 sent successfully in 0.9638276100158691 seconds.
first half iteration 8 sent successfully in 19.77590584754944 seconds.
second half iteration 8 sent successfully in 6.07660174369812 seconds.
*************************************************************************
first half avg: sent successfully in 4.763573974370956 seconds.
second half avg: sent successfully in 2.4090722799301147 seconds.
total avg: sent successfully in 6.792858645319939 seconds.
*************************************************************************
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

25% loss:

Sending file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc change dev lo root netem loss 25%
[sudo] password for jodogo:
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
Connected to 127.0.0.1:9999
Recived ack on file size
Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n)
```

Receiving file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
Connected by ('127.0.0.1', 49482)
first half recived - size of : 3145463
seoond half recived - size of : 6287941
file recived
```

Finished sending file multiple times to the receiver:

```
total file sent - size of : 6291456
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n) n
Sent exit message to receiver.
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

Receiver prints stats: here we can see the average time is now above one minute, we see a awful outliner last line of 437 seconds and conclude that the difference between 10 - 15% is not the same as 20 - 25% we start to conclude exponential time in regards to level of loss %.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL

file recived
Connected by ('127.0.0.1', 49482)
first half recived - size of : 3144367
seoond half recived - size of : 6291066
file recived
Connected by ('127.0.0.1', 49482)
first half recived - size of : 3143184
seoond half recived - size of : 6289989
file recived

Statistics:
*************************************************************************
first half iteration 1 sent successfully in 3.324810266494751 seconds.
second half iteration 1 sent successfully in 31.39530110359192 seconds.
first half iteration 2 sent successfully in 0.9808380603790283 seconds.
second half iteration 2 sent successfully in 18.945563793182373 seconds.
first half iteration 3 sent successfully in 99.49193859100342 seconds.
second half iteration 3 sent successfully in 4.474587917327881 seconds.
first half iteration 4 sent successfully in 2.3464291095733643 seconds.
second half iteration 4 sent successfully in 0.9952306747436523 seconds.
first half iteration 5 sent successfully in 4.9648237228393555 seconds.
second half iteration 5 sent successfully in 437.05942583084106 seconds.
*************************************************************************
first half avg: sent successfully in 22.221767950057984 seconds.
second half avg: sent successfully in 98.57402186393738 seconds.
total avg: sent successfully in 77.08984723091126 seconds.
*************************************************************************
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

30% loss:

Sent file for the first time:

```
● jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc change dev lo root netem loss 30%
  [sudo] password for jodogo:
○ jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
  Connected to 127.0.0.1:9999
  Recived ack on file size
  Sent first half of file:
  Authentication received for first half: b'1000111000010110001010'
  CC algorithm changed to cubic:
  Sent second half of file:
  Authentication received for second half: b'1000111000010110001010'
  Sent second half of file:

  total file sent - size of : 2097152
  Do you wan
```

Received file for the first time:

```
○ jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
  Connected by ('127.0.0.1', 51164)
  first half recived - size of : 1044480
  seoond half recived - size of : 2096781
  file recived
```

Finished sending file multiple times (5 times as this was taking forever):

```
  Sent first half of file:
  Authentication received for first half: b'1000111000010110001010'
  CC algorithm changed to cubic:
  Sent second half of file:
  Authentication received for second half: b'1000111000010110001010'
  Sent second half of file:

  total file sent - size of : 2097152
  Do you want to send the file again? (y/n) y
  Notified receiver to send again.
  CC algorithm changed back to reno.

  Sent first half of file:
  Authentication received for first half: b'1000111000010110001010'
  CC algorithm changed to cubic:
  Sent second half of file:
  Authentication received for second half: b'1000111000010110001010'
  Sent second half of file:

  total file sent - size of : 2097152
  Do you want to send the file again? (y/n) n
  Sent exit message to receiver.
  jodogo@jod
  ogo-Virtua
  lBox:~/Des
  ktop/Matal
○ jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

Receiver finished receiving the last time the file was sent and prints stats:

```
first half recived - size of : 1047834
seoond half recived - size of : 2095545
file recived
Connected by ('127.0.0.1', 51164)
first half recived - size of : 1047728
seoond half recived - size of : 2095386
file recived

Statistics:
*************************************************************************
first half iteration 1 sent successfully in 1.3683812618255615 seconds.
second half iteration 1 sent successfully in 8.704511880874634 seconds.
first half iteration 2 sent successfully in 1.8507919311523438 seconds.
second half iteration 2 sent successfully in 13.570802450180054 seconds.
first half iteration 3 sent successfully in 5.474904298782349 seconds.
second half iteration 3 sent successfully in 2.3657453060150146 seconds.
first half iteration 4 sent successfully in 177.15200638771057 seconds.
second half iteration 4 sent successfully in 16.435860633850098 seconds.
first half iteration 5 sent successfully in 2.018491506576538 seconds.
second half iteration 5 sent successfully in 50.11139893531799 seconds.
first half iteration 6 sent successfully in 1330.7247593402863 seconds.
second half iteration 6 sent successfully in 630.7837238311768 seconds.
*************************************************************************
first half avg: sent successfully in 253.09822245438895 seconds.
second half avg: sent successfully in 120.32867383956909 seconds.
total avg: sent successfully in 320.8615859746933 seconds.
*************************************************************************
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

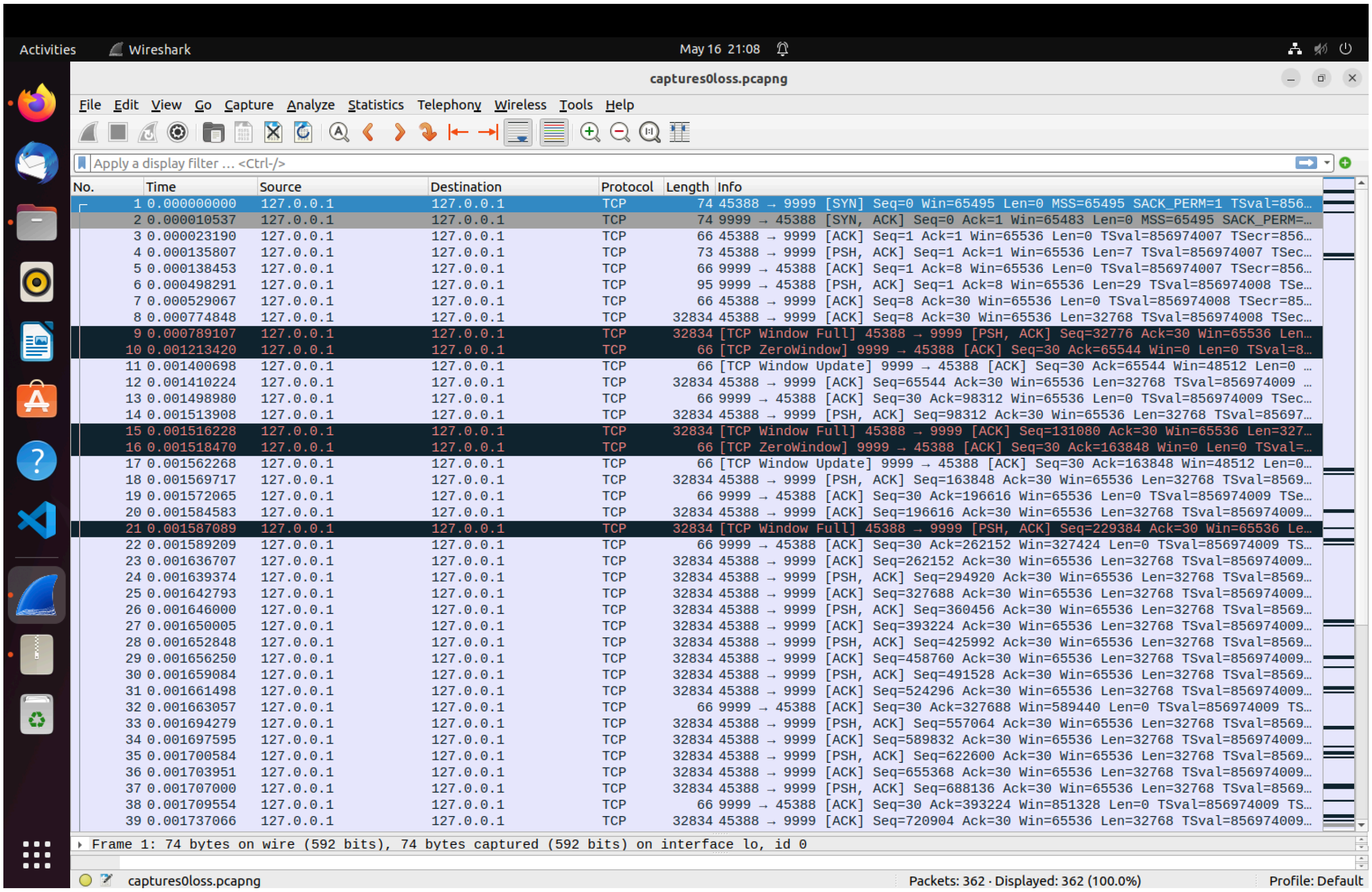
Key Findings:

- We set the receivers wait time to 2*(longest time a half was sent) to be careful as with high % level of packet loss the receiver would close to quickly. This then meant that the sender would close gracefully and the receiver would wait along time before printing the stats and then also exiting gracfully.
- We found that the time is exponential and the last rounds of loss level % 25 - 30 were much much much longer as opposed to the 10-15% levels wich were bearable.
- We used flags to ensure the TCP connection was only made once and stayed open until the sender decided to exit.
- We noticed outliners in the data that would effectively affect the averages in regards to times being printed by the receiver.
- We were receiving errors when closing the receiver connection and server and printed outputs to debug where things were going wrong. By using recursion we concluded that in a while loop listening and receiving data from the sender, that if we closed the connection and server of the receiver within a recursion call we had to make sure when it comes out of recursion that it exits gracefully, meaning that when using recursion to be careful that all resources are still available to use and if not to make a flag and exit gracefully.
- Our code would work With out loss of generality to the data being sent and decided to send a file of text.

Wireshark recordings:

in order to make it easier on the reader, in each packet loss screenshot, the connection between the sender and receiver will be at the top row in the first picture.

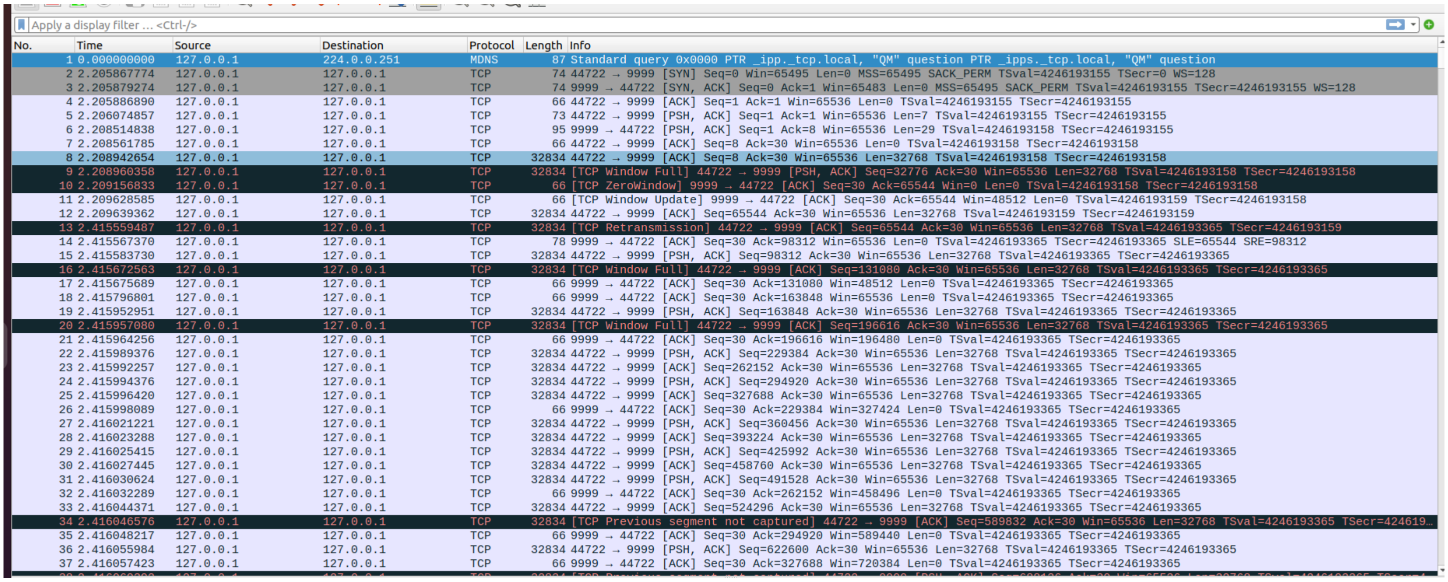
0% loss:

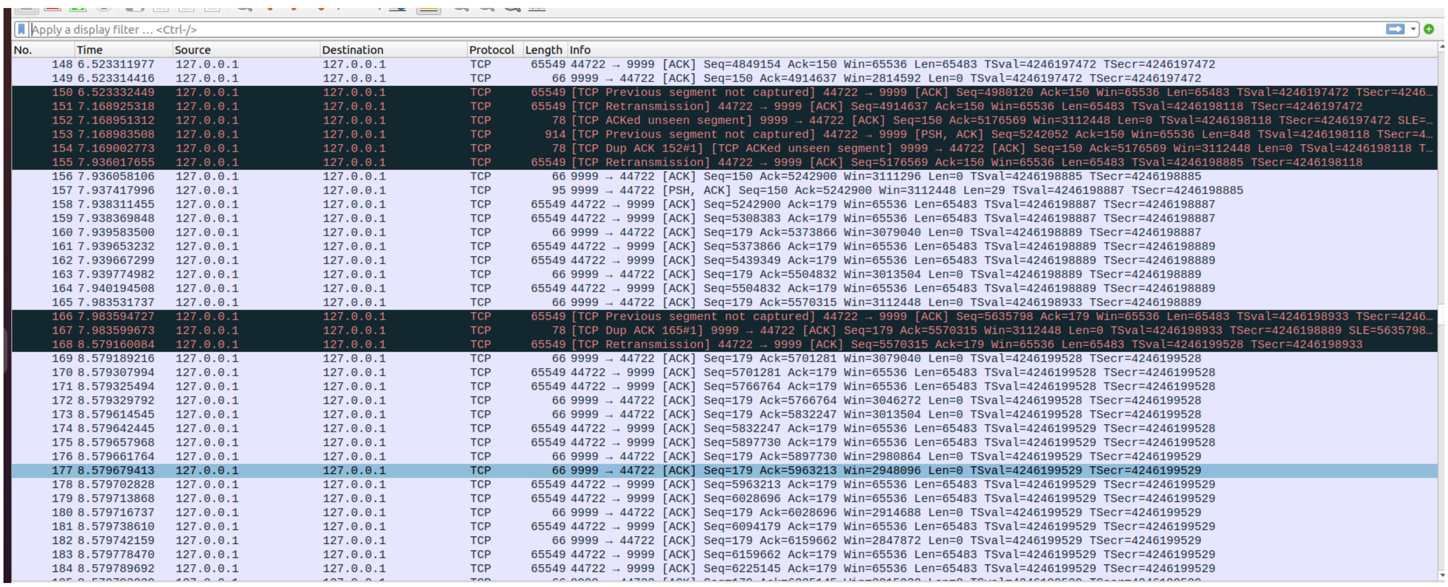


The "[TCP Window Full]" message from Wireshark means that the system sending this TCP segment has filled up the receive window of the other end with the tcp segment in this packet.

Zero Window means that the receiver of the packets telling the sender to stop sending because there is no more buffer space for incoming packets. This is in almost all cases a sign of the receiver being too slow to process the incoming packets in time.

10% loss:





Lost packets:

"tcp previous segment not captured" is a message created by Wireshark when it didn't see a packet that should have been in the trace.

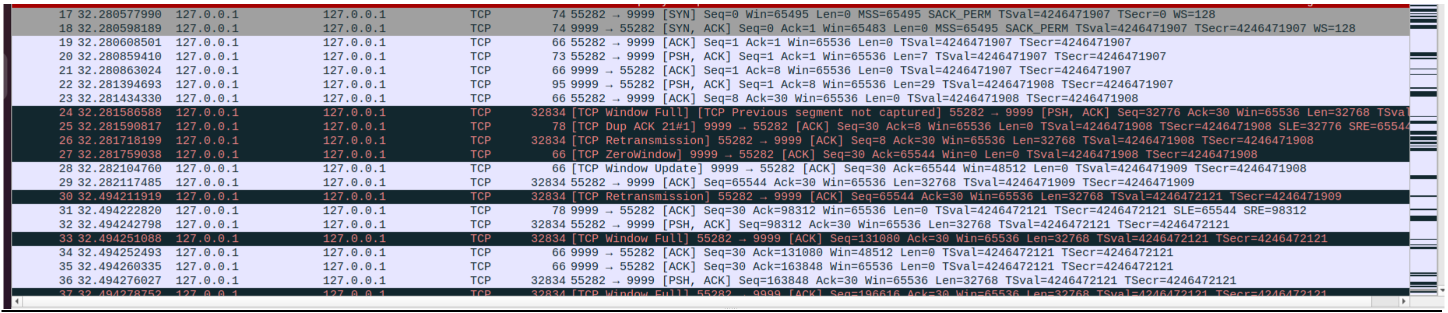
Dup ACK means that you captured at the source of the data (not the receiving side). That is quite normal if the packet loss occurs somewhere in the path to the receiver.

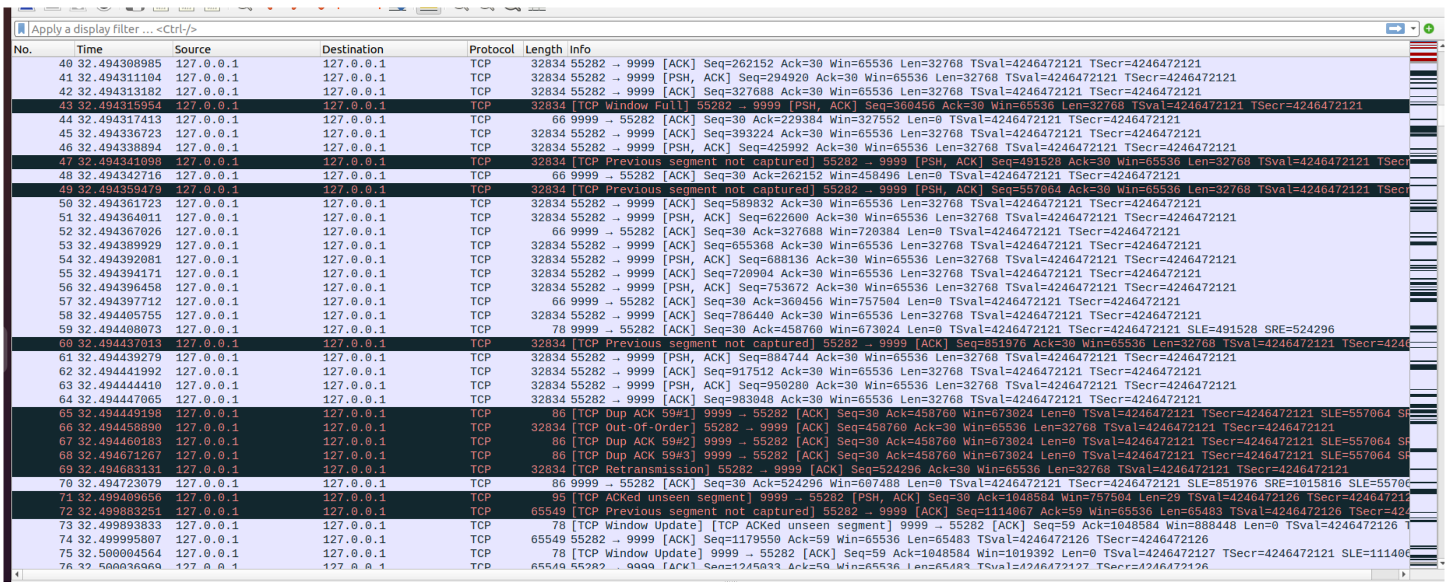
TCP out-of-order means that particular frame was received in a different order from which it was sent (after a later packet in the sequence). It probably indicates that there are multiple paths between source and destination - and one travels a through a longer path.

TCP Retransmission occurs when the sender retransmits a packet after the expiration of the acknowledgement.

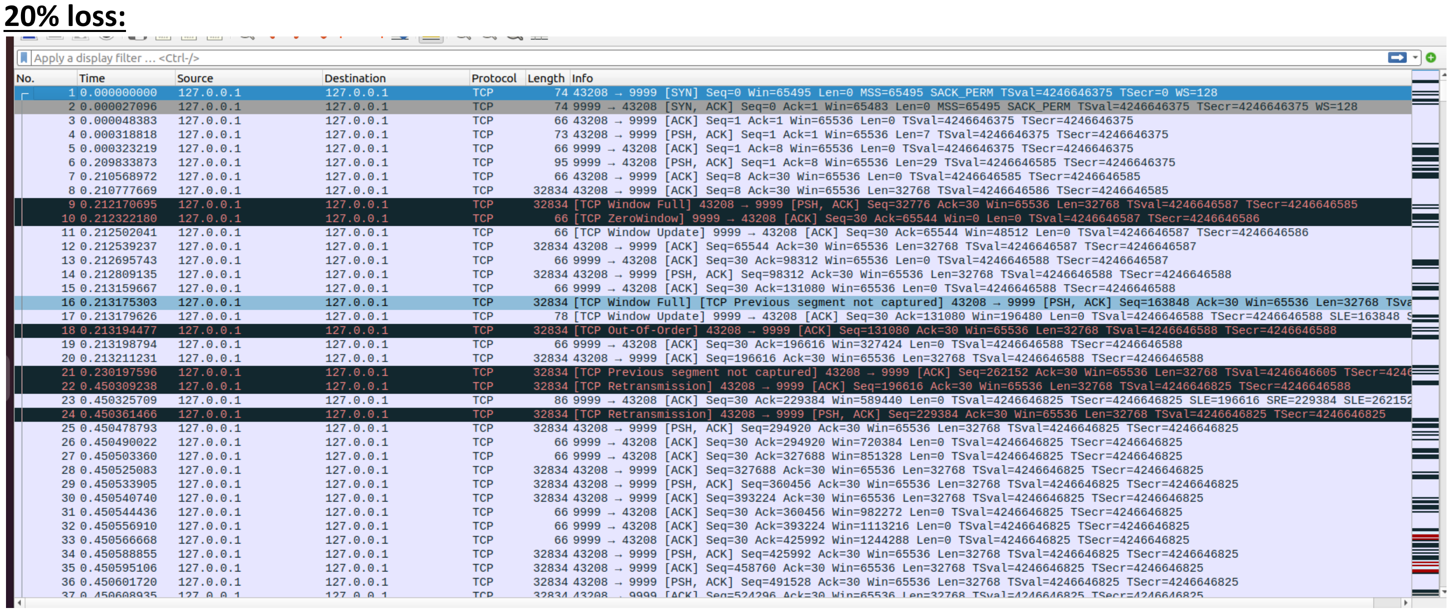
"TCP ACKed unseen segment" means that this packet acknowledges data that wasn't captured. It was transferred okay, and the receiver acknowledges it, but Wireshark can't find the packet in the capture. This usually happens when the capture device wasn't fast enough.

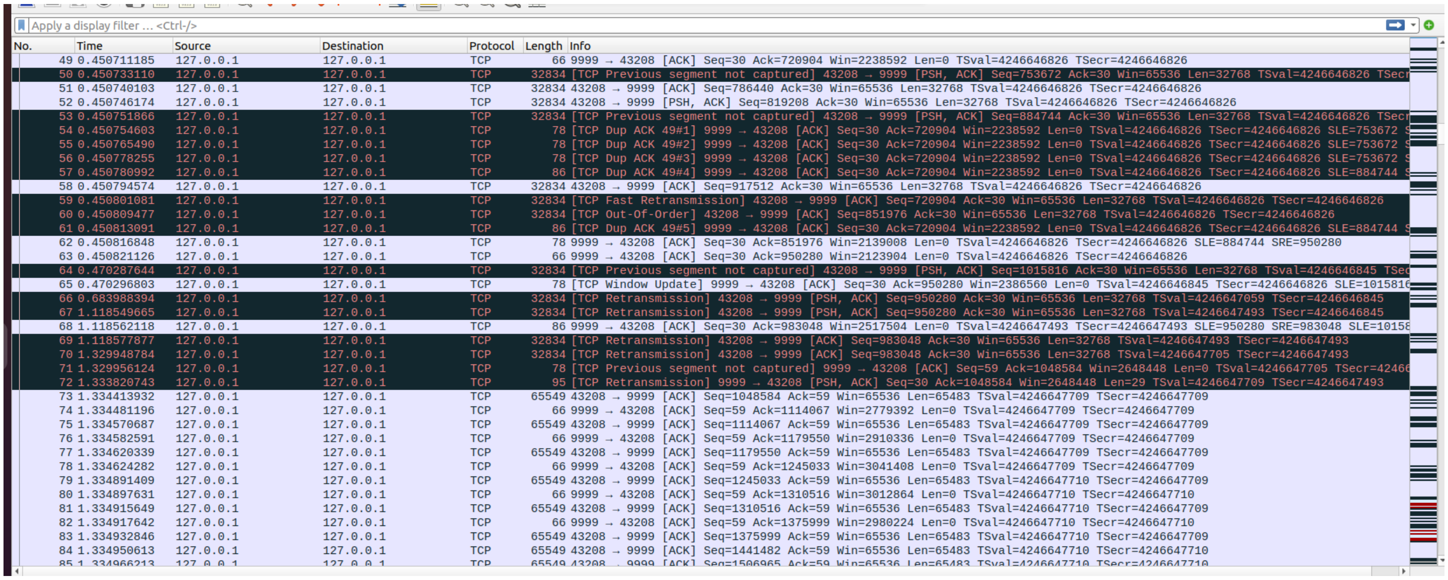
15% loss:



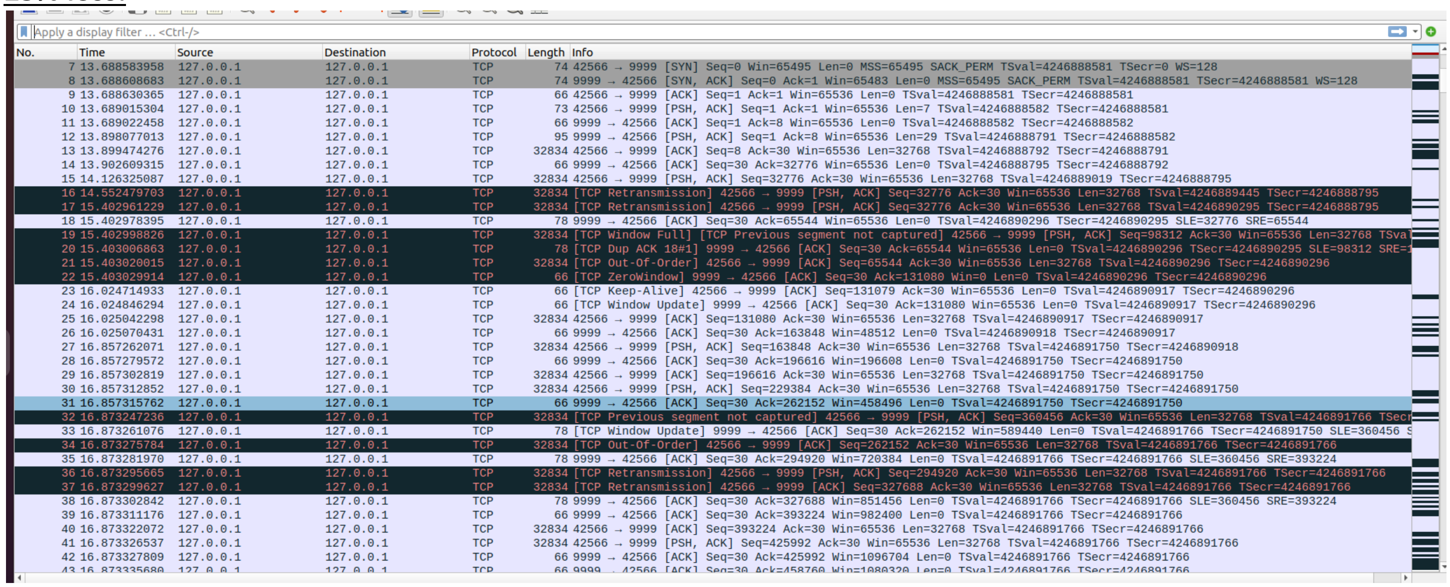


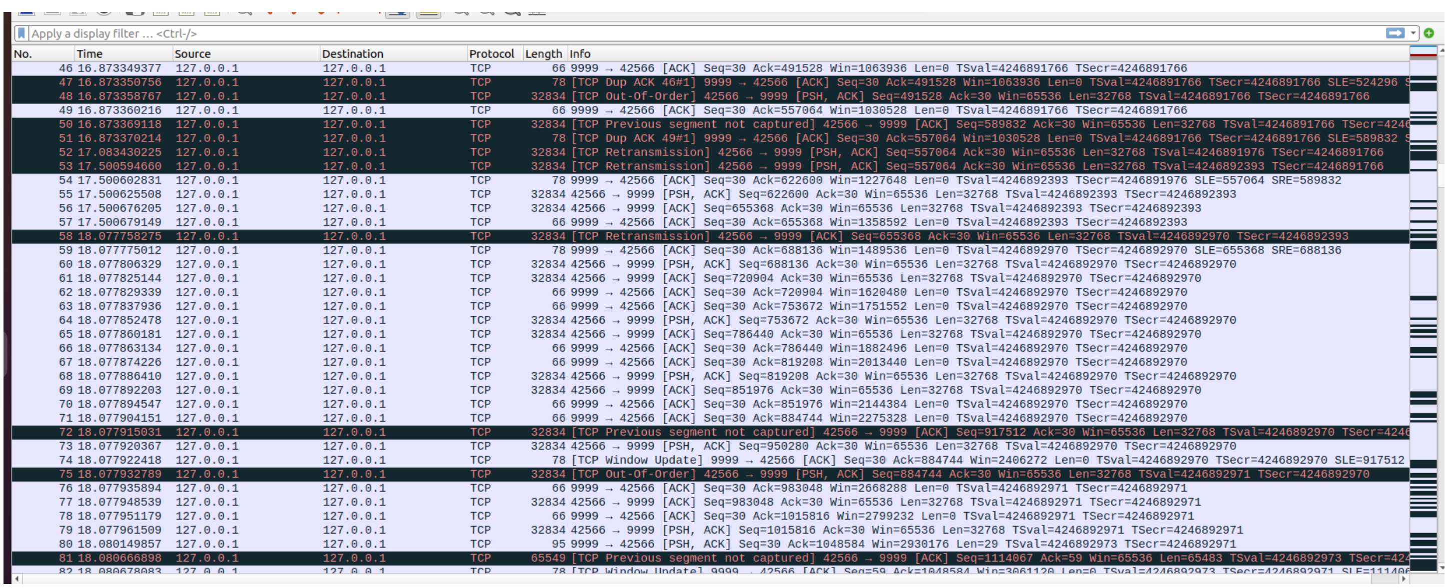
20% loss:




Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
49 0.450711185 127.0.0.1 127.0.0.1 TCP 66 9999 → 43208 [ACK] Seq=30 Ack=720904 Win=2238592 Len=0 TSval=4246646826 TSecr=4246646826
50 0.450733110 127.0.0.1 127.0.0.1 TCP 32834 [TCP Previous segment not captured] 43208 → 9999 [PSH, ACK] Seq=753672 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr
51 0.450740103 127.0.0.1 127.0.0.1 TCP 32834 43208 → 9999 [ACK] Seq=786440 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr=4246646826
52 0.450746174 127.0.0.1 127.0.0.1 TCP 32834 43208 → 9999 [PSH, ACK] Seq=819208 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr=4246646826
53 0.450751866 127.0.0.1 127.0.0.1 TCP 32834 [TCP Previous segment not captured] 43208 → 9999 [PSH, ACK] Seq=884744 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr
54 0.450754603 127.0.0.1 127.0.0.1 TCP 78 [TCP Dup ACK 49#1] 9999 → 43208 [ACK] Seq=30 Ack=720904 Win=2238592 Len=0 TSval=4246646826 TSecr=4246646826 SLE=753672
55 0.450765490 127.0.0.1 127.0.0.1 TCP 78 [TCP Dup ACK 49#2] 9999 → 43208 [ACK] Seq=30 Ack=720904 Win=2238592 Len=0 TSval=4246646826 TSecr=4246646826 SLE=753672
56 0.450778255 127.0.0.1 127.0.0.1 TCP 78 [TCP Dup ACK 49#3] 9999 → 43208 [ACK] Seq=30 Ack=720904 Win=2238592 Len=0 TSval=4246646826 TSecr=4246646826 SLE=753672
57 0.450780992 127.0.0.1 127.0.0.1 TCP 86 [TCP Dup ACK 49#4] 9999 → 43208 [ACK] Seq=30 Ack=720904 Win=2238592 Len=0 TSval=4246646826 TSecr=4246646826 SLE=884744
58 0.450794574 127.0.0.1 127.0.0.1 TCP 32834 43208 → 9999 [ACK] Seq=917512 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr=4246646826
59 0.450801081 127.0.0.1 127.0.0.1 TCP 32834 [TCP Fast Retransmission] 43208 → 9999 [ACK] Seq=720904 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr=4246646826
60 0.450809477 127.0.0.1 127.0.0.1 TCP 32834 [TCP Out-Of-Order] 43208 → 9999 [ACK] Seq=851976 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr=4246646826
61 0.450813091 127.0.0.1 127.0.0.1 TCP 86 [TCP Dup ACK 49#5] 9999 → 43208 [ACK] Seq=30 Ack=720904 Win=2238592 Len=0 TSval=4246646826 TSecr=4246646826 SLE=884744
62 0.450816848 127.0.0.1 127.0.0.1 TCP 78 9999 → 43208 [ACK] Seq=30 Ack=851976 Win=2139008 Len=0 TSval=4246646826 TSecr=4246646826 SLE=884744 SRE=950280
63 0.450821126 127.0.0.1 127.0.0.1 TCP 66 9999 → 43208 [ACK] Seq=30 Ack=950280 Win=2123904 Len=0 TSval=4246646826 TSecr=4246646826
64 0.470287644 127.0.0.1 127.0.0.1 TCP 32834 [TCP Previous segment not captured] 43208 → 9999 [PSH, ACK] Seq=1015816 Ack=30 Win=65536 Len=32768 TSval=4246646845 TSec
65 0.470296803 127.0.0.1 127.0.0.1 TCP 78 [TCP Window Update] 9999 → 43208 [ACK] Seq=30 Ack=950280 Win=2386560 Len=0 TSval=4246646845 TSecr=4246646845 SLE=101581
66 0.683988394 127.0.0.1 127.0.0.1 TCP 32834 [TCP Retransmission] 43208 → 9999 [PSH, ACK] Seq=950280 Ack=30 Win=65536 Len=32768 TSval=4246647059 TSecr=4246646845
67 1.118549665 127.0.0.1 127.0.0.1 TCP 32834 [TCP Retransmission] 43208 → 9999 [PSH, ACK] Seq=950280 Ack=30 Win=65536 Len=32768 TSval=4246647493 TSecr=4246646845
68 1.118562118 127.0.0.1 127.0.0.1 TCP 86 9999 → 43208 [ACK] Seq=30 Ack=983048 Win=2517504 Len=0 TSval=4246647493 TSecr=4246647493 SLE=950280 SRE=983048 SLE=10158
69 1.118577877 127.0.0.1 127.0.0.1 TCP 32834 [TCP Retransmission] 43208 → 9999 [ACK] Seq=983048 Ack=30 Win=65536 Len=32768 TSval=4246647493 TSecr=4246647493
70 1.329948784 127.0.0.1 127.0.0.1 TCP 32834 [TCP Retransmission] 43208 → 9999 [ACK] Seq=983048 Ack=30 Win=65536 Len=32768 TSval=4246647705 TSecr=4246647493
71 1.329956124 127.0.0.1 127.0.0.1 TCP 78 [TCP Previous segment not captured] 9999 → 43208 [ACK] Seq=59 Ack=1048584 Win=2648448 Len=0 TSval=4246647705 TSecr=42466
72 1.333820743 127.0.0.1 127.0.0.1 TCP 95 [TCP Retransmission] 9999 → 43208 [PSH, ACK] Seq=30 Ack=1048584 Win=2648448 Len=29 TSval=4246647709 TSecr=4246647493
73 1.334413932 127.0.0.1 127.0.0.1 TCP 65549 43208 → 9999 [ACK] Seq=1048584 Ack=59 Win=65536 Len=65483 TSval=4246647709 TSecr=4246647709
74 1.334481196 127.0.0.1 127.0.0.1 TCP 66 9999 → 43208 [ACK] Seq=59 Ack=1114067 Win=2779392 Len=0 TSval=4246647709 TSecr=4246647709
75 1.334570687 127.0.0.1 127.0.0.1 TCP 65549 43208 → 9999 [ACK] Seq=1114067 Ack=59 Win=65536 Len=65483 TSval=4246647709 TSecr=4246647709
76 1.334582591 127.0.0.1 127.0.0.1 TCP 66 9999 → 43208 [ACK] Seq=59 Ack=1179550 Win=2910336 Len=0 TSval=4246647709 TSecr=4246647709
77 1.334620339 127.0.0.1 127.0.0.1 TCP 65549 43208 → 9999 [ACK] Seq=1179550 Ack=59 Win=65536 Len=65483 TSval=4246647709 TSecr=4246647709
78 1.334624282 127.0.0.1 127.0.0.1 TCP 66 9999 → 43208 [ACK] Seq=59 Ack=1245033 Win=3041408 Len=0 TSval=4246647709 TSecr=4246647709
79 1.334891409 127.0.0.1 127.0.0.1 TCP 65549 43208 → 9999 [ACK] Seq=1245033 Ack=59 Win=65536 Len=65483 TSval=4246647710 TSecr=4246647709
80 1.334897631 127.0.0.1 127.0.0.1 TCP 66 9999 → 43208 [ACK] Seq=59 Ack=1310516 Win=3012864 Len=0 TSval=4246647710 TSecr=4246647710
81 1.334915649 127.0.0.1 127.0.0.1 TCP 65549 43208 → 9999 [ACK] Seq=1310516 Ack=59 Win=65536 Len=65483 TSval=4246647710 TSecr=4246647709
82 1.334917642 127.0.0.1 127.0.0.1 TCP 66 9999 → 43208 [ACK] Seq=59 Ack=1375999 Win=2980224 Len=0 TSval=4246647710 TSecr=4246647710
83 1.334932846 127.0.0.1 127.0.0.1 TCP 65549 43208 → 9999 [ACK] Seq=1375999 Ack=59 Win=65536 Len=65483 TSval=4246647710 TSecr=4246647709
84 1.334950613 127.0.0.1 127.0.0.1 TCP 65549 43208 → 9999 [ACK] Seq=1441482 Ack=59 Win=65536 Len=65483 TSval=4246647710 TSecr=4246647710

25% loss:


Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
7 13.688583958 127.0.0.1 127.0.0.1 TCP 74 42566 → 9999 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=4246888581 TSecr=0 WS=128
8 13.688608683 127.0.0.1 127.0.0.1 TCP 74 9999 → 42566 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=4246888581 TSecr=4246888581 WS=128
9 13.688630365 127.0.0.1 127.0.0.1 TCP 66 42566 → 9999 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=4246888581 TSecr=4246888581
10 13.689015304 127.0.0.1 127.0.0.1 TCP 73 42566 → 9999 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=7 TSval=4246888582 TSecr=4246888581
11 13.689022458 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=1 Ack=8 Win=65536 Len=0 TSval=4246888582 TSecr=4246888582
12 13.898077013 127.0.0.1 127.0.0.1 TCP 95 9999 → 42566 [PSH, ACK] Seq=1 Ack=8 Win=65536 Len=29 TSval=4246888791 TSecr=4246888582
13 13.899474276 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=32768 TSval=4246888792 TSecr=4246888791
14 13.902609315 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=32776 Win=65536 Len=0 TSval=4246888795 TSecr=4246888792
15 14.126325087 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [PSH, ACK] Seq=32776 Ack=30 Win=65536 Len=32768 TSval=4246889019 TSecr=4246888795
16 14.552479703 127.0.0.1 127.0.0.1 TCP 32834 [TCP Retransmission] 42566 → 9999 [PSH, ACK] Seq=32776 Ack=30 Win=65536 Len=32768 TSval=4246889445 TSecr=4246888795
17 15.402961229 127.0.0.1 127.0.0.1 TCP 32834 [TCP Retransmission] 42566 → 9999 [PSH, ACK] Seq=32776 Ack=30 Win=65536 Len=32768 TSval=4246890295 TSecr=4246888795
18 15.402978395 127.0.0.1 127.0.0.1 TCP 78 9999 → 42566 [ACK] Seq=30 Ack=65544 Win=65536 Len=0 TSval=4246890296 TSecr=4246890295 SLE=32776 SRE=65544
19 15.402998826 127.0.0.1 127.0.0.1 TCP 32834 [TCP Window Full] [TCP Previous segment not captured] 42566 → 9999 [PSH, ACK] Seq=98312 Ack=30 Win=65536 Len=32768 TSva
20 15.403006863 127.0.0.1 127.0.0.1 TCP 78 [TCP Dup ACK 18#1] 9999 → 42566 [ACK] Seq=30 Ack=65544 Win=65536 Len=0 TSval=4246890296 TSecr=4246890295 SLE=98312 SRE=
21 15.403020015 127.0.0.1 127.0.0.1 TCP 32834 [TCP Out-Of-Order] 42566 → 9999 [ACK] Seq=65544 Ack=30 Win=65536 Len=32768 TSval=4246890296 TSecr=4246890296
22 15.403029914 127.0.0.1 127.0.0.1 TCP 66 [TCP ZeroWindow] 9999 → 42566 [ACK] Seq=30 Ack=131080 Win=0 Len=0 TSval=4246890296 TSecr=4246890296
23 16.024714933 127.0.0.1 127.0.0.1 TCP 66 [TCP Keep-Alive] 42566 → 9999 [ACK] Seq=131079 Ack=30 Win=65536 Len=0 TSval=4246890917 TSecr=4246890296
24 16.024846294 127.0.0.1 127.0.0.1 TCP 66 [TCP Window Update] 9999 → 42566 [ACK] Seq=30 Ack=131080 Win=65536 Len=0 TSval=4246890917 TSecr=4246890296
25 16.025042298 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [ACK] Seq=131080 Ack=30 Win=65536 Len=32768 TSval=4246890917 TSecr=4246890917
26 16.025070431 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=163848 Win=48512 Len=0 TSval=4246890918 TSecr=4246890917
27 16.857262071 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [PSH, ACK] Seq=163848 Ack=30 Win=65536 Len=32768 TSval=4246891750 TSecr=4246890918
28 16.857279572 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=196616 Win=196608 Len=0 TSval=4246891750 TSecr=4246891750
29 16.857302819 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [ACK] Seq=196616 Ack=30 Win=65536 Len=32768 TSval=4246891750 TSecr=4246891750
30 16.857312852 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [PSH, ACK] Seq=229384 Ack=30 Win=65536 Len=32768 TSval=4246891750 TSecr=4246891750
31 16.857315762 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=262152 Win=458496 Len=0 TSval=4246891750 TSecr=4246891750
32 16.873247236 127.0.0.1 127.0.0.1 TCP 32834 [TCP Previous segment not captured] 42566 → 9999 [PSH, ACK] Seq=360456 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr
33 16.873261076 127.0.0.1 127.0.0.1 TCP 78 [TCP Window Update] 9999 → 42566 [ACK] Seq=30 Ack=262152 Win=589440 Len=0 TSval=4246891766 TSecr=4246891750 SLE=360456
34 16.873275784 127.0.0.1 127.0.0.1 TCP 32834 [TCP Out-Of-Order] 42566 → 9999 [ACK] Seq=262152 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
35 16.873281970 127.0.0.1 127.0.0.1 TCP 78 9999 → 42566 [ACK] Seq=30 Ack=294920 Win=720384 Len=0 TSval=4246891766 TSecr=4246891766 SLE=360456 SRE=393224
36 16.873295665 127.0.0.1 127.0.0.1 TCP 32834 [TCP Retransmission] 42566 → 9999 [PSH, ACK] Seq=294920 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
37 16.873299627 127.0.0.1 127.0.0.1 TCP 32834 [TCP Retransmission] 42566 → 9999 [ACK] Seq=327688 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
38 16.873302842 127.0.0.1 127.0.0.1 TCP 78 9999 → 42566 [ACK] Seq=30 Ack=327688 Win=851456 Len=0 TSval=4246891766 TSecr=4246891766 SLE=360456 SRE=393224
39 16.873311176 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=393224 Win=982400 Len=0 TSval=4246891766 TSecr=4246891766
40 16.873322072 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [ACK] Seq=393224 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
41 16.873326537 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [PSH, ACK] Seq=425992 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
42 16.873327809 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=425992 Win=1096704 Len=0 TSval=4246891766 TSecr=4246891766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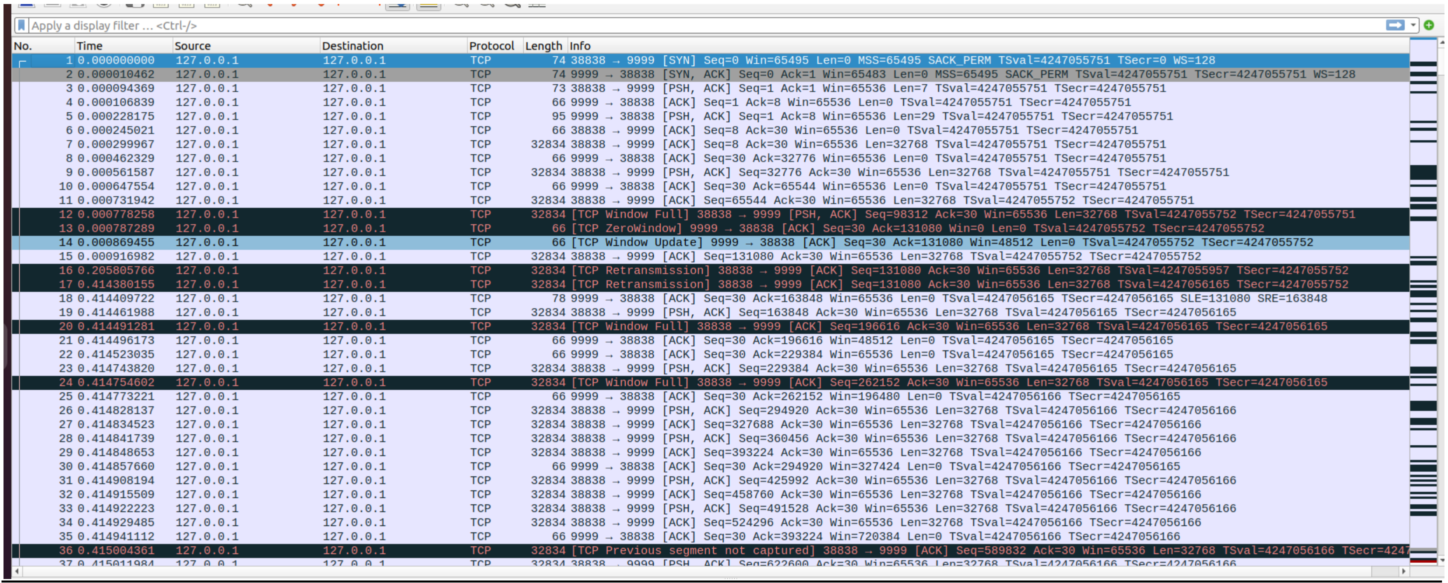
No. Time Source Destination Protocol Length Info
46 16.873349377 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=491528 Win=1063936 Len=0 TSval=4246891766 TSecr=4246891766
47 16.873350756 127.0.0.1 127.0.0.1 TCP 78 [TCP Dup ACK 46#1] 9999 → 42566 [ACK] Seq=30 Ack=491528 Win=1063936 Len=0 TSval=4246891766 TSecr=4246891766 SLE=524296
48 16.873358767 127.0.0.1 127.0.0.1 TCP 32834 [TCP Out-Of-Order] 42566 → 9999 [PSH, ACK] Seq=491528 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
49 16.873360216 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=557064 Win=1030528 Len=0 TSval=4246891766 TSecr=4246891766
50 16.873369118 127.0.0.1 127.0.0.1 TCP 32834 [TCP Previous segment not captured] 42566 → 9999 [ACK] Seq=589832 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=424
51 16.873370214 127.0.0.1 127.0.0.1 TCP 78 [TCP Dup ACK 49#1] 9999 → 42566 [ACK] Seq=30 Ack=557064 Win=1030528 Len=0 TSval=4246891766 TSecr=4246891766 SLE=589832
52 17.083430225 127.0.0.1 127.0.0.1 TCP 32834 [TCP Retransmission] 42566 → 9999 [PSH, ACK] Seq=557064 Ack=30 Win=65536 Len=32768 TSval=4246891976 TSecr=4246891766
53 17.500594660 127.0.0.1 127.0.0.1 TCP 32834 [TCP Retransmission] 42566 → 9999 [PSH, ACK] Seq=557064 Ack=30 Win=65536 Len=32768 TSval=4246892393 TSecr=4246891766
54 17.500602831 127.0.0.1 127.0.0.1 TCP 78 9999 → 42566 [ACK] Seq=30 Ack=622600 Win=1227648 Len=0 TSval=4246892393 TSecr=4246891976 SLE=557064 SRE=589832
55 17.500625508 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [PSH, ACK] Seq=622600 Ack=30 Win=65536 Len=32768 TSval=4246892393 TSecr=4246892393
56 17.500676205 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [ACK] Seq=655368 Ack=30 Win=65536 Len=32768 TSval=4246892393 TSecr=4246892393
57 17.500679149 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=655368 Win=1358592 Len=0 TSval=4246892393 TSecr=4246892393
58 18.077758275 127.0.0.1 127.0.0.1 TCP 32834 [TCP Retransmission] 42566 → 9999 [ACK] Seq=655368 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892393
59 18.077775012 127.0.0.1 127.0.0.1 TCP 78 9999 → 42566 [ACK] Seq=30 Ack=688136 Win=1489536 Len=0 TSval=4246892970 TSecr=4246892970 SLE=655368 SRE=688136
60 18.077806329 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [PSH, ACK] Seq=688136 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
61 18.077825144 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [ACK] Seq=720904 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
62 18.077829339 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=720904 Win=1620480 Len=0 TSval=4246892970 TSecr=4246892970
63 18.077837936 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=753672 Win=1751552 Len=0 TSval=4246892970 TSecr=4246892970
64 18.077852478 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [PSH, ACK] Seq=753672 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
65 18.077860181 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [ACK] Seq=786440 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
66 18.077863134 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=786440 Win=1882496 Len=0 TSval=4246892970 TSecr=4246892970
67 18.077874226 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=819208 Win=2013440 Len=0 TSval=4246892970 TSecr=4246892970
68 18.077886410 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [PSH, ACK] Seq=819208 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
69 18.077892203 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [ACK] Seq=851976 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
70 18.077894547 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=851976 Win=2144384 Len=0 TSval=4246892970 TSecr=4246892970
71 18.077904151 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=884744 Win=2275328 Len=0 TSval=4246892970 TSecr=4246892970
72 18.077915031 127.0.0.1 127.0.0.1 TCP 32834 [TCP Previous segment not captured] 42566 → 9999 [ACK] Seq=917512 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=424
73 18.077920367 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [PSH, ACK] Seq=950280 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
74 18.077922418 127.0.0.1 127.0.0.1 TCP 78 [TCP Window Update] 9999 → 42566 [ACK] Seq=30 Ack=884744 Win=2406272 Len=0 TSval=4246892970 TSecr=4246892970 SLE=917512
75 18.077932789 127.0.0.1 127.0.0.1 TCP 32834 [TCP Out-Of-Order] 42566 → 9999 [PSH, ACK] Seq=884744 Ack=30 Win=65536 Len=32768 TSval=4246892971 TSecr=4246892970
76 18.077935894 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=983048 Win=2668288 Len=0 TSval=4246892971 TSecr=4246892971
77 18.077948539 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [ACK] Seq=983048 Ack=30 Win=65536 Len=32768 TSval=4246892971 TSecr=4246892971
78 18.077951179 127.0.0.1 127.0.0.1 TCP 66 9999 → 42566 [ACK] Seq=30 Ack=1015816 Win=2799232 Len=0 TSval=4246892971 TSecr=4246892971
79 18.077961509 127.0.0.1 127.0.0.1 TCP 32834 42566 → 9999 [PSH, ACK] Seq=1015816 Ack=30 Win=65536 Len=32768 TSval=4246892971 TSecr=4246892971
80 18.080149857 127.0.0.1 127.0.0.1 TCP 95 9999 → 42566 [PSH, ACK] Seq=30 Ack=1048584 Win=2930176 Len=29 TSval=4246892973 TSecr=4246892971
81 18.080666898 127.0.0.1 127.0.0.1 TCP 65549 [TCP Previous segment not captured] 42566 → 9999 [ACK] Seq=1114067 Ack=59 Win=65536 Len=65483 TSval=4246892973 TSecr=424

30% loss:



Apply a display filter … <Ctrl-/>

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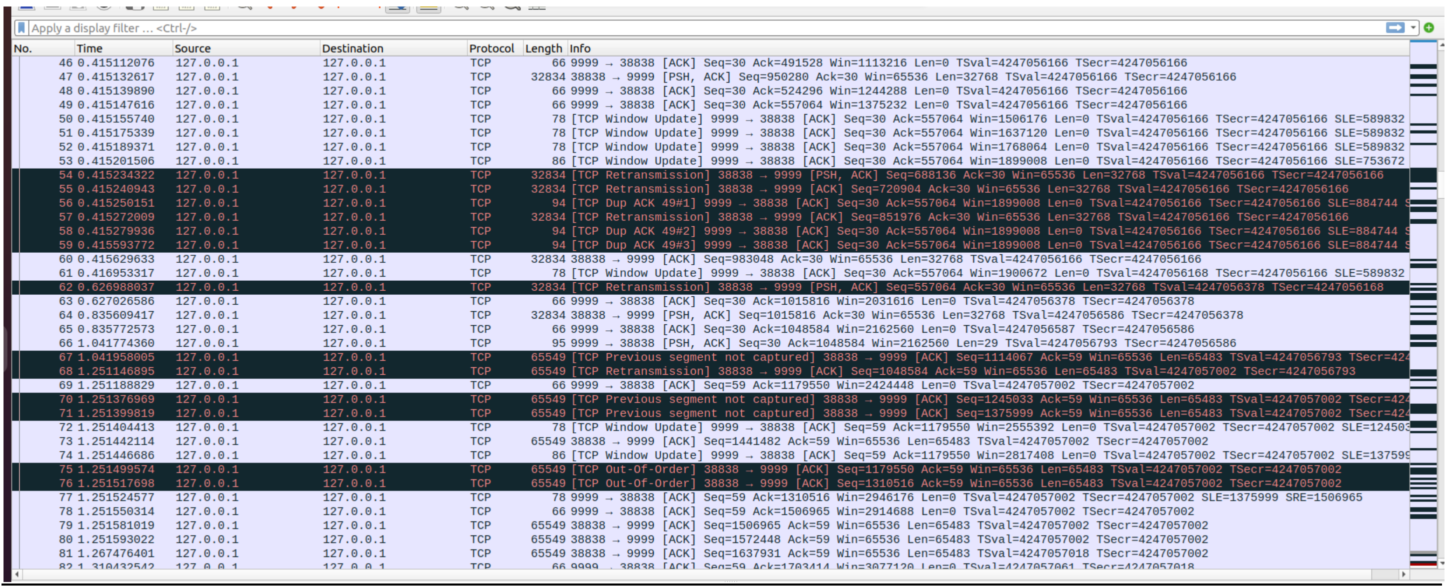
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	38838 → 9999 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=4247055751 TSecr=0 WS=128
2	0.000010462	127.0.0.1	127.0.0.1	TCP	74	9999 → 38838 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=4247055751 TSecr=4247055751 WS=128
3	0.000094369	127.0.0.1	127.0.0.1	TCP	73	38838 → 9999 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=7 TSval=4247055751 TSecr=4247055751
4	0.000106839	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=1 Ack=8 Win=65536 Len=0 TSval=4247055751 TSecr=4247055751
5	0.000228175	127.0.0.1	127.0.0.1	TCP	95	9999 → 38838 [PSH, ACK] Seq=1 Ack=8 Win=65536 Len=29 TSval=4247055751 TSecr=4247055751
6	0.000245021	127.0.0.1	127.0.0.1	TCP	66	38838 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=0 TSval=4247055751 TSecr=4247055751
7	0.000299967	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=32768 TSval=4247055751 TSecr=4247055751
8	0.000462329	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=32776 Win=65536 Len=0 TSval=4247055751 TSecr=4247055751
9	0.000561587	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=32776 Ack=30 Win=65536 Len=32768 TSval=4247055751 TSecr=4247055751
10	0.000647554	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=65544 Win=65536 Len=0 TSval=4247055751 TSecr=4247055751
11	0.000731942	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=65544 Ack=30 Win=65536 Len=32768 TSval=4247055752 TSecr=4247055751
12	0.000778258	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 38838 → 9999 [PSH, ACK] Seq=98312 Ack=30 Win=65536 Len=32768 TSval=4247055752 TSecr=4247055751
13	0.000787289	127.0.0.1	127.0.0.1	TCP	66	[TCP ZeroWindow] 9999 → 38838 [ACK] Seq=30 Ack=131080 Win=0 Len=0 TSval=4247055752 TSecr=4247055752
14	0.000869455	127.0.0.1	127.0.0.1	TCP	66	[TCP Window Update] 9999 → 38838 [ACK] Seq=30 Ack=131080 Win=48512 Len=0 TSval=4247055752 TSecr=4247055752
15	0.000916982	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=131080 Ack=30 Win=65536 Len=32768 TSval=4247055752 TSecr=4247055752
16	0.205805766	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 38838 → 9999 [ACK] Seq=131080 Ack=30 Win=65536 Len=32768 TSval=4247055957 TSecr=4247055752
17	0.414380155	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 38838 → 9999 [ACK] Seq=131080 Ack=30 Win=65536 Len=32768 TSval=4247056165 TSecr=4247055752
18	0.414409722	127.0.0.1	127.0.0.1	TCP	78	9999 → 38838 [ACK] Seq=30 Ack=163848 Win=65536 Len=0 TSval=4247056165 TSecr=4247056165 SLE=131080 SRE=163848
19	0.414461988	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=163848 Ack=30 Win=65536 Len=32768 TSval=4247056165 TSecr=4247056165
20	0.414491281	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 38838 → 9999 [ACK] Seq=196616 Ack=30 Win=65536 Len=32768 TSval=4247056165 TSecr=4247056165
21	0.414496173	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=196616 Win=48512 Len=0 TSval=4247056165 TSecr=4247056165
22	0.414523035	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=229384 Win=65536 Len=0 TSval=4247056165 TSecr=4247056165
23	0.414743820	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=229384 Ack=30 Win=65536 Len=32768 TSval=4247056165 TSecr=4247056165
24	0.414754602	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 38838 → 9999 [ACK] Seq=262152 Ack=30 Win=65536 Len=32768 TSval=4247056165 TSecr=4247056165
25	0.414773221	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=262152 Win=196480 Len=0 TSval=4247056166 TSecr=4247056165
26	0.414828137	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=294920 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
27	0.414834523	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=327688 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
28	0.414841739	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=360456 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
29	0.414848653	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=393224 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
30	0.414857660	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=294920 Win=327424 Len=0 TSval=4247056166 TSecr=4247056165
31	0.414908194	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=425992 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
32	0.414915509	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=458760 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
33	0.414922223	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=491528 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
34	0.414929485	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=524296 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
35	0.414941112	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=393224 Win=720384 Len=0 TSval=4247056166 TSecr=4247056166
36	0.415004361	127.0.0.1	127.0.0.1	TCP	32834	[TCP Previous segment not captured] 38838 → 9999 [ACK] Seq=589832 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247
37	0.415011984	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=622600 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166



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No.	Time	Source	Destination	Protocol	Length	Info
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47	0.415132617	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=950280 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
48	0.415139890	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=524296 Win=1244288 Len=0 TSval=4247056166 TSecr=4247056166
49	0.415147616	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1375232 Len=0 TSval=4247056166 TSecr=4247056166
50	0.415155740	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1506176 Len=0 TSval=4247056166 TSecr=4247056166 SLE=589832
51	0.415175339	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1637120 Len=0 TSval=4247056166 TSecr=4247056166 SLE=589832
52	0.415189371	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1768064 Len=0 TSval=4247056166 TSecr=4247056166 SLE=589832
53	0.415201506	127.0.0.1	127.0.0.1	TCP	86	[TCP Window Update] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1899008 Len=0 TSval=4247056166 TSecr=4247056166 SLE=753672
54	0.415234322	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 38838 → 9999 [PSH, ACK] Seq=688136 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
55	0.415240943	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 38838 → 9999 [ACK] Seq=720904 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
56	0.415250151	127.0.0.1	127.0.0.1	TCP	94	[TCP Dup ACK 49#1] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1899008 Len=0 TSval=4247056166 TSecr=4247056166 SLE=884744 S
57	0.415272009	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 38838 → 9999 [ACK] Seq=851976 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
58	0.415279936	127.0.0.1	127.0.0.1	TCP	94	[TCP Dup ACK 49#2] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1899008 Len=0 TSval=4247056166 TSecr=4247056166 SLE=884744 S
59	0.415593772	127.0.0.1	127.0.0.1	TCP	94	[TCP Dup ACK 49#3] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1899008 Len=0 TSval=4247056166 TSecr=4247056166 SLE=884744 S
60	0.415629633	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=983048 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
61	0.416953317	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1900672 Len=0 TSval=4247056168 TSecr=4247056166 SLE=589832
62	0.626988037	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 38838 → 9999 [PSH, ACK] Seq=557064 Ack=30 Win=65536 Len=32768 TSval=4247056378 TSecr=4247056168
63	0.627026586	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=1015816 Win=2031616 Len=0 TSval=4247056378 TSecr=4247056378
64	0.835609417	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=1015816 Ack=30 Win=65536 Len=32768 TSval=4247056586 TSecr=4247056378
65	0.835772573	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=1048584 Win=2162560 Len=0 TSval=4247056587 TSecr=4247056586
66	1.041774360	127.0.0.1	127.0.0.1	TCP	95	9999 → 38838 [PSH, ACK] Seq=30 Ack=1048584 Win=2162560 Len=29 TSval=4247056793 TSecr=4247056586
67	1.041958005	127.0.0.1	127.0.0.1	TCP	65549	[TCP Previous segment not captured] 38838 → 9999 [ACK] Seq=1114067 Ack=59 Win=65536 Len=65483 TSval=4247056793 TSecr=424
68	1.251146895	127.0.0.1	127.0.0.1	TCP	65549	[TCP Retransmission] 38838 → 9999 [ACK] Seq=1048584 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247056793
69	1.251188829	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=59 Ack=1179550 Win=2424448 Len=0 TSval=4247057002 TSecr=4247057002
70	1.251376969	127.0.0.1	127.0.0.1	TCP	65549	[TCP Previous segment not captured] 38838 → 9999 [ACK] Seq=1245033 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=424
71	1.251399819	127.0.0.1	127.0.0.1	TCP	65549	[TCP Previous segment not captured] 38838 → 9999 [ACK] Seq=1375999 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=424
72	1.251404413	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 38838 [ACK] Seq=59 Ack=1179550 Win=2555392 Len=0 TSval=4247057002 TSecr=4247057002 SLE=124503
73	1.251442114	127.0.0.1	127.0.0.1	TCP	65549	38838 → 9999 [ACK] Seq=1441482 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247057002
74	1.251446686	127.0.0.1	127.0.0.1	TCP	86	[TCP Window Update] 9999 → 38838 [ACK] Seq=59 Ack=1179550 Win=2817408 Len=0 TSval=4247057002 TSecr=4247057002 SLE=137599
75	1.251499574	127.0.0.1	127.0.0.1	TCP	65549	[TCP Out-Of-Order] 38838 → 9999 [ACK] Seq=1179550 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247057002
76	1.251517698	127.0.0.1	127.0.0.1	TCP	65549	[TCP Out-Of-Order] 38838 → 9999 [ACK] Seq=1310516 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247057002
77	1.251524577	127.0.0.1	127.0.0.1	TCP	78	9999 → 38838 [ACK] Seq=59 Ack=1310516 Win=2946176 Len=0 TSval=4247057002 TSecr=4247057002 SLE=1375999 SRE=1506965
78	1.251550314	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=59 Ack=1506965 Win=2914688 Len=0 TSval=4247057002 TSecr=4247057002
79	1.251581019	127.0.0.1	127.0.0.1	TCP	65549	38838 → 9999 [ACK] Seq=1506965 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247057002
80	1.251593022	127.0.0.1	127.0.0.1	TCP	65549	38838 → 9999 [ACK] Seq=1572448 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247057002
81	1.267476401	127.0.0.1	127.0.0.1	TCP	65549	38838 → 9999 [ACK] Seq=1637931 Ack=59 Win=65536 Len=65483 TSval=4247057018 TSecr=4247057002
82	1.310432542	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=59 Ack=1703414 Win=3077120 Len=0 TSval=4247057061 TSecr=4247057018